

PROJECT TOTAL WORK ALLOCATION

PROJECT TITLE:

**PhishShield: A Machine Learning-
Based Web Application for
Phishing URL Detection**

GROUP MEMBERS:

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Introduction:

With the rapid growth of online banking, e-commerce, and digital communication, phishing attacks have become one of the most dangerous cyber threats. Cybercriminals create fake websites that closely resemble legitimate ones to deceive users into revealing confidential information. Since phishing websites can be created quickly, traditional blacklist-based methods often fail to detect newly generated malicious URLs.

Machine learning provides an intelligent approach by analyzing the characteristics of a URL and identifying suspicious patterns. This project develops a web application that allows users to verify the authenticity of URLs before accessing them, thereby improving online safety.

TEAM WORK DISTRIBUTION:

The project is divided among four members based on their technical expertise. Each member is responsible for a specific module while collaborating with the team during integration and testing.

Member 1 – Front-End Developer

Responsibilities:

- Design an attractive and responsive user interface.
- Develop the home page and phishing detection page.
- Create input forms for URL submission.
- Display prediction results in a user-friendly format.
- Handle user interactions and client-side validation.
- Ensure compatibility across different browsers.

Technologies Used:

- HTML5
- CSS3
- JavaScript
- Bootstrap (Optional)

Deliverables:

- Responsive web interface
- URL input page
- Result display page
- Navigation and user experience improvements

Member 2 – Back-End Developer::

Responsibilities:

- Develop the server-side application.
- Receive URL requests from the frontend.
- Connect the frontend with the machine learning model.
- Handle API requests and responses.
- Manage the project database (if required).
- Implement authentication (future enhancement).

Technologies Used:

- Python
- Flask
- REST API
- MySQL (Optional)

Deliverables:

- Backend server
- API endpoints
- Database connectivity
- Model integration

Member 3 – Machine Learning Engineer

Responsibilities:

- Collect phishing and legitimate URL datasets.
- Clean and preprocess the dataset.
- Extract important URL features.
- Train and evaluate machine learning models.
- Select the best-performing model.
- Export the trained model for deployment.

Machine Learning Tasks:

- Data Collection
- Data Cleaning
- Feature Engineering
- Model Training
- Hyperparameter Tuning
- Performance Evaluation

Algorithms (Example):

- Random Forest
- XGBoost
- Logistic Regression
- Support Vector Machine

Deliverables:

- Clean dataset
- Trained ML model
- Accuracy report
- Saved prediction model

Member 4 – Cybersecurity & Testing Engineer

Responsibilities:

- Study phishing attack techniques.
- Validate the effectiveness of the detection system.
- Perform functional testing.
- Conduct security testing against malicious URLs.
- Verify application performance and reliability.
- Document bugs and improvements.

Testing Activities:

- Unit Testing
- Integration Testing
- Functional Testing
- Security Testing
- Performance Testing
- User Acceptance Testing (UAT)

Deliverables:

- Test cases
- Security assessment report
- Bug reports
- Final testing documentation

Collaboration Between **Members**

The team will work collaboratively throughout the project lifecycle

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- **Front-End ↔ Back-End: Integration of the user interface with server APIs.**
- **Back-End ↔ Machine Learning: Deployment of the trained ML model into the web application.**
- **Machine Learning ↔ Cybersecurity: Validation of prediction accuracy using phishing datasets.**
- **Cybersecurity ↔ Entire Team: Continuous testing, bug reporting, and quality assurance.**

Summary:

This work allocation ensures that all four members contribute equally while focusing on distinct aspects of the project. The Front-End Developer is responsible for creating an intuitive user interface, the Back-End Developer manages server-side functionality and integration, the Machine Learning Engineer develops the phishing detection model, and the Cybersecurity & Testing Engineer validates the system's security, reliability, and overall performance. Regular collaboration among all members will ensure successful integration and timely completion of the project.

REFERENCES:

- R. Verma and A. Das, "What's in a URL: Fast Feature Extraction and Malicious URL Detection," Proceedings of the ACM Conference on Information and Knowledge Management, 2017.
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- S. Garera et al., "A Framework for Detection and Measurement of Phishing Attacks," Proceedings of the ACM Workshop on Recurring Malcode, 2007.
- J. Ma et al., "Beyond Blacklists: Learning to Detect Malicious Web Sites from Suspicious URLs," Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2009.